



Exchange & Capital Markets AI Use Cases

Use case cards, prioritization methodology, and implementation guidance

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1. Executive Summary

Capital markets participants are deploying AI across trading, settlement, risk management, and compliance. This guide catalogs 15+ proven use cases with implementation patterns, ROI models, and prioritization frameworks.

2. Market Overview

Market Growth: AI/ML in capital markets projected to grow 25% CAGR through 2030 **Investment Areas:** Trading automation, settlement optimization, fraud detection, regulatory technology **Key Drivers:** Cost reduction, speed, scalability, competitive differentiation

3. Use Case Categories

Trading & Execution

- Smart order routing: Route orders to optimal venues
- Price prediction: Forecast short-term price movements
- Algorithmic trading: Adaptive execution algorithms
- Portfolio optimization: Dynamic rebalancing

Risk & Compliance

- Fraud detection: Real-time transaction monitoring
- Counterparty risk: Predict default probability
- Regulatory reporting: Automated data extraction and validation
- Sanctions screening: Enhanced AML/KYC screening

Operations & Settlement

- Settlement exception handling: Automated resolution
- Reconciliation: Automated matching and investigation
- Cash optimization: Forecast and optimize liquidity
- Cybersecurity: Threat detection and response

4. Use Case Cards

Use Case: Smart Order Routing

Dimension	Details
Description	AI system routes orders to optimal trading venues based on real-time market data
Business Impact	\$500K-2M annual savings (reduced execution costs, faster fills)
Implementation Time	6-9 months
Data Required	Order flow, venue prices, liquidity, latency data

Key Metrics	% fills at best price, average execution time, cost per trade
Risk Level	Medium (market impact, system failures)
Regulatory	MiFID II best execution, SEC Reg SHO

Use Case: Real-Time Fraud Detection

Dimension	Details
Description	ML models identify anomalous trading patterns indicating potential fraud or market abuse
Business Impact	\$1-5M annual savings (prevented losses, regulatory penalties avoided)
Implementation Time	3-6 months
Data Required	Trade data, order flow, account profiles, settlement instructions
Key Metrics	Detection rate, false positive rate, time to alert, investigation time
Risk Level	Medium (false positives, detection gaps)
Regulatory	GDPR, Reg SHO, MAR, MiFID II

5. Prioritization Framework

Use Case Scoring Matrix

Criterion	Weight	Scoring	Notes
Financial Impact	30%	High (>\$1M): 3 Medium: 2 Low: 1	Revenue or savings
Implementation Complexity	20%	Low (3-6 mo): 3 Medium: 2 High: 1	Lower is better
Data Readiness	20%	Ready: 3 Partial: 2 Not Ready: 1	Data availability
Regulatory Risk	15%	Low: 3 Medium: 2 High: 1	Compliance burden
Strategic Alignment	15%	High: 3 Medium: 2 Low: 1	Fits strategy

6. Implementation Guidance

Phase 1: Pilot (3-4 months)

- Select specific instrument or market segment
- Build proof-of-concept with backtest
- Establish monitoring and controls
- Paper trade or small volume production test

Phase 2: Limited Production (2-3 months)

- Deploy to 10-20% of target volume
- Monitor performance, ROI, risks
- Gather feedback, refine models
- Establish incident response procedures

Phase 3: Full Deployment (1-2 months)

- Scale to full volume
- Transition to operational team
- Document procedures and governance
- Begin continuous optimization

7. ROI Models

Smart Order Routing ROI

Year	Investment	Cost Savings	Revenue Impact	Net Benefit
Year 1	\$500K	\$750K	\$200K	\$450K
Year 2	\$150K	\$1.2M	\$500K	\$1.55M
Year 3	\$100K	\$1.2M	\$800K	\$1.9M

Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

We're here to help.

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For consultation on implementation, governance, and strategic deployment of AI initiatives.

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